

Acetaminophen hepatotoxicity and acute liver failure.

PMID: 19169150 | Indexed in PubMed

Acetaminophen-induced hepatotoxicity is a common consequence of acetaminophen overdose and may lead to acute liver failure (ALF). Currently acetaminophen is the most common cause of ALF in both United States and United Kingdom, with a trend to increasing incidence in the United States. N-acetylcysteine is the most effective drug to prevent progression to liver failure with acetaminophen hepatotoxicity. Liver transplantation is the only definitive therapy that will significantly increase the chances of survival for advanced ALF. This communication reviews current information regarding causes and management of acetaminophen-induced hepatotoxicity and ALF.

Randomised open label exploratory, safety and tolerability study with calmagafodipir in patients treated with the 12-h regimen of N-acetylcysteine for paracetamol overdose-the PP100-01 for Overdose of Paracetamol (POP) trial: study protocol for a randomised controlled trial.

PMID: 30621764 | Indexed in PubMed

Paracetamol (acetaminophen) overdose (POD) is the commonest cause of acute liver failure in Europe and North America. Current treatment involves the use of the antidote N-acetylcysteine (NAC) in patients deemed at risk of liver damage. This regimen was introduced in the 1970s and has remained largely unchanged even though the initial NAC infusion is frequently associated with adverse reactions, in particular nausea, vomiting, and anaphylactoid reactions. NAC has reduced efficacy for preventing liver injury in those patients who present later after overdose. We designed a randomised study investigating the safety and tolerability of a superoxide dismutase (SOD) mimetic, calmagafodipir (PP100-01), co-treatment with a 12-h NAC regimen compared with NAC treatment alone in patients with POD. We have designed an open-label, randomised, exploratory, rising dose design, NAC-controlled, phase 1 safety and tolerability study in patients treated with NAC for POD. A total of 24 patients will be assigned into one of three dosing cohorts of eight patients ($n = 6$ for PP100-01 and NAC; $n = 2$ for NAC alone). The doses of PP100-01 are 2, 5, and 10 $\mu\text{mol/kg}$. The primary outcome is the safety and tolerability of PP100-01 when co-administered with a 12-h NAC regimen compared with NAC treatment alone. Furthermore, the study will explore if PP100-01 has potential efficacy for the treatment of paracetamol-induced liver injury by measurement of conventional clinical and exploratory biomarkers. The aim of the study is to test the safety and tolerability of a SOD mimetic, PP100-01, in combination with a 12-h NAC regimen in patients presenting within 24 h of POD. This study will provide valuable data regarding the incidence of adverse events caused by the 12-h NAC plus PP100-01 regimen and may provide evidence of PP100-01 efficacy in the treatment of paracetamol-induced liver injury. EudraCT, 2017-000246-21; ClinicalTrials.gov, NCT03177395 . Registered on 6 June 2017.

Acute liver failure including acetaminophen overdose.

PMID: 18570942 | Indexed in PubMed

Acute liver failure (ALF) is a dramatic, highly unpredictable clinical syndrome defined by the sudden onset of coagulopathy and encephalopathy. Acetaminophen overdose, the leading cause of ALF in the

United States, has a 66% chance of recovery with early N-acetylcysteine treatment and supportive care. Cerebral edema and infectious complications are difficult to detect and treat in these patients and may cause irreversible brain damage and multiorgan failure. One-year survival after emergency liver transplantation is 70%, but 20% of listed patients die, highlighting the importance of early referral of patients who have ALF with a poor prognosis to a transplant center.

Improvement by acetylcysteine of hemodynamics and oxygen transport in fulminant hepatic failure.

PMID: 1904133 | Indexed in PubMed

When administered early after an overdose of acetaminophen, intravenous acetylcysteine prevents hepatic necrosis by replenishing reduced stores of glutathione. How acetylcysteine improves the survival of patients with established liver damage induced by acetaminophen, however, is unknown. This study was undertaken to determine whether the beneficial effect of acetylcysteine under such circumstances could be due to enhancement of oxygen delivery and consumption. We studied the effect of acetylcysteine on systemic hemodynamics and oxygen transport in 12 patients with acetaminophen-induced fulminant hepatic failure and 8 patients with acute liver failure from other causes. The acetylcysteine was given in a dose of 150 mg per kilogram of body weight in 250 ml of 5 percent dextrose over a period of 15 minutes and then in a dose of 50 mg per kilogram in 500 ml of 5 percent dextrose over a period of 4 hours; measurements were made before treatment began and after 30 minutes of the regimen. In the patients with acetaminophen-induced liver failure, the infusion of acetylcysteine resulted in an increase in mean oxygen delivery from 856 to 975 ml per minute per square meter of body-surface area ($P = 0.0036$), due to an increase in the cardiac index from 5.6 to 6.7 liters per minute per square meter ($P = 0.0021$). Mean arterial pressure rose from 88 to 95 mm Hg ($P = 0.0054$) despite a decrease in systemic vascular resistance from 1296 to 1113 dyn.sec.cm⁻⁵ per square meter ($P = 0.027$). There was an increase in oxygen consumption from 127 to 184 ml per minute per square meter ($P = 0.0007$) associated with an increase in the oxygen-extraction ratio from 16 to 21 percent ($P = 0.022$). The effects in the patients with acute liver failure from other causes were similar. The increase in oxygen delivery and consumption in response to acetylcysteine may account for its beneficial effect on survival in patients with fulminant hepatic failure induced by acetaminophen.

Acetaminophen (APAP or N-Acetyl-p-Aminophenol) and Acute Liver Failure.

PMID: 29605069 | Indexed in PubMed

Acetaminophen (APAP) is the leading cause of acute liver failure (ALF), although the worldwide frequency is variable. APAP hepatotoxicity develops either following intentional overdose or unintentional ingestion (therapeutic misadventure) in the background of several factors, such as concomitant use of alcohol and certain medications that facilitate the formation of reactive and toxic metabolites. Spontaneous survival is more common in APAP-induced ALF compared with non-APAP etiologies. N-acetylcysteine is recommended for all patients with APAP-induced ALF and it reduces mortality. Liver transplantation should be offered early to those who are unlikely to survive based on described prognostic criteria.

Late N-acetylcysteine for successful recovery of acetaminophen-related acute liver failure: A case report.

PMID: 37767143 | Indexed in PubMed

Acetaminophen toxicity is one of the leading causes of liver failure. Although N-acetylcysteine (NAC) is generally successful in preventing acetaminophen hepatotoxicity when given in a timely manner, if not prescribed in the early golden time, the only practical way to save the patient might be liver transplantation. The case presented was a 20-year-old female with an acetaminophen overdose (30 g), for which more than 24 h had passed since the ingestion. Despite the critical clinical condition, loss of consciousness (Glasgow Coma Score of 4) of the patient, and passing the golden time of antidote administration, the decision was made by the healthcare team to administer NAC. After transferring the patient to the intensive care unit, the three-bag NAC regimen was initiated and appropriate monitoring was performed. After this, the regimen of 3 g q8h was continued for the patient. The patient's condition began to improve slowly on the second day and then she was extubated on the fourth day. Finally, she was discharged on the tenth day. Although the golden period of antidote administration had passed outwardly, there was no need for a liver transplant and the patient recovered successfully with late NAC administration. Hence, clinicians can benefit from the use of NAC even in the late phases of acetaminophen liver toxicity.

Acute Liver Failure.

PMID: 15527712 | Indexed in PubMed

Acute liver failure (ALF) is an uncommon medical emergency whose rapid progression and high mortality demand early diagnosis and expert management, including immediate transfer of any potential case to facilities for intensive care and orthotopic liver transplantation (OLT). All patients with ALF must be screened aggressively for acetaminophen toxicity (history, serum levels, "hyperacute" presentation with renal failure), for other drugs, and viral hepatitis; rare causes of ALF should also be considered. After an acetaminophen overdose, N-acetylcysteine must be given as early as possible, preferably in the emergency room, but any patient with ALF should promptly receive N-acetylcysteine if there is suspicion of acetaminophen toxicity irrespective of the time of ingestion. Supportive care for all patients with ALF includes adequate enteral nutrition, aggressive screening and treatment of infection, prophylactic broad-spectrum antibiotics, and antifungal agents. Sedation with propofol is given for severe agitation or mechanical ventilation. With advanced coma grades, intensive care is needed with hemodynamic monitoring, ventilatory support, continuous renal replacement for renal failure, and intracranial pressure monitoring. Intracranial hypertension is treated with mannitol and/or acute short-term hyperventilation, but if the patient is refractory to treatment, mild-moderate hypothermia is achieved by a cooling blanket that is continued throughout OLT. Barbiturate coma is only used in refractory cases as the last treatment modality. Seizures are aggressively treated with phenytoin, with additional diazepam as needed. Candidacy and activation for OLT should be completed as early as possible in the course of ALF, especially in "hyperacute" cases such as acetaminophen toxicity. The final decision to proceed with OLT is made when a donor organ becomes available. King's College Hospital criteria for OLT are still the best prognostic assessment for fatal outcome in ALF, but the criteria fail to identify some patients who will die.

[Severe toxic acute liver failure: etiology and treatment].

PMID: 23683460 | Indexed in PubMed

Many substances, drugs or not, can be responsible for acute hepatitis. Nevertheless, toxic etiology, except when that is obvious like in acetaminophen overdose, is a diagnosis of elimination. Major causes, in particular viral etiologies, must be ruled out. Acetaminophen, antibiotics, antiepileptics and antituberculous drugs are the first causes of drug-induced liver injury. Severity assessment of the acute hepatitis is critical. Acute liver failure (ALF) is defined by the factor V, respectively more than 50% for the mild ALF and less than 50% for the severe ALF. Neurological examination must be extensive to the search for encephalopathy signs. According to the French classification, fulminant hepatitis is defined by the presence of an encephalopathy in the two first weeks and subfulminant between the second and 12th week after the advent of the jaundice. During acetaminophen overdose, with or without hepatitis or ALF, intravenous N-acetylcysteine must be administered as soon as possible. In the non-acetaminophen related ALF, N-acetylcysteine improves transplantation-free survival. Referral and assessment in a liver transplantation unit should be discussed as soon as possible.

N-acetylcysteine increases cerebral perfusion pressure in pigs with fulminant hepatic failure.

PMID: 11588469 | Indexed in PubMed

Intravenous administration of N-acetylcysteine beyond 15 hrs reduces mortality rates in patients suffering from paracetamol-induced fulminant hepatic failure, although the mechanism of the therapeutic benefit remains unclear. We hypothesized increased survival to be caused by improved hemodynamic performance. The main objective for the study was to explore the effect of N-acetylcysteine on hemodynamics, oxygen transport, and regional blood flow in pigs with fulminant hepatic failure. Prospective, randomized, controlled trial. Surgical research laboratory in a university hospital. Female Norwegian Landrace pigs. Fulminant hepatic failure was induced by a total liver devascularization procedure. Five hours later, the pigs were allocated to N-acetylcysteine treatment (150 mg.kg⁻¹ in 100 mL of 0.9% saline over 15 mins, followed by 50 mg.kg⁻¹ in 500 mL of 0.9% saline over a period of 4 hrs) or placebo. Mean arterial pressure stabilized in the N-acetylcysteine group and increased slightly during the last 2 hrs (pGT =.009). Thus, mean arterial pressure was significantly higher compared with placebo after 3 hrs (p =.01). Cerebral perfusion pressure was significantly higher during the last 2 hrs in the N-acetylcysteine group (pGT =.033). Common carotid artery flow also increased and was maintained at a higher level compared with placebo (pG =.027). Systemic vascular resistance index initially decreased but then gradually increased (pGT <.001). Cardiac index increased after 15 mins of N-acetylcysteine infusion, causing a significant interaction (pGT =.038), but did not differ after 3 hrs. No significant differences in hindleg and mesentery hemodynamics were found. A short-lived increase in oxygen delivery caused by a temporary increase in cardiac index was observed but without any corresponding increase in oxygen consumption. Intravenous N-acetylcysteine infusion increases cerebral perfusion pressure in pigs with fulminant hepatic failure. Earlier reported effects on oxygen transport and uptake could not be confirmed.

Epidemiology of acute liver failure.

PMID: 17335680 | Indexed in PubMed

Acute liver failure (ALF) is an uncommon disorder that leads to jaundice, coagulopathy, and multisystem organ failure. Its definition is based on the timing from onset of jaundice to encephalopathy. In 2005, ALF accounted for 6% of liver-related deaths and 7% of orthotopic liver transplants (OLT) in the United States. Several classification systems have been developed for ALF, with the King's College criteria most widely used for prediction of OLT. Specific diagnostic tests should be implemented to identify the cause of ALF, which will help to determine its treatment and prognosis. Viral hepatitis was previously reported to be the most common cause of ALF in the United States, but acetaminophen overdose and idiosyncratic drug reactions have emerged as the most frequent causes in recent studies. Malignancy is an uncommon cause of ALF, and thus imaging studies may not be useful in this setting, but liver biopsy may be beneficial in selected cases. An overall strategy for ALF should start with identifying the cause, assessing the prognosis, and early transfer to a transplantation center for suitable candidates. OLT has emerged as a life-saving procedure leading to marked improvement in survival rates. Improved surgical techniques, immunosuppression, and comprehensive care have led to an overall survival rate of approximately 65% with OLT. N-acetylcysteine is effective in ALF caused by acetaminophen overdose, with results strongly related to how soon it is given rather than the route of administration. Liver support systems show potential for the treatment of ALF, but their role needs validation in large multicenter randomized trials.
